RHC Paediatric Intensive Care Unit

Making alterations to Plasmalyte IV maintenance bags

Plasmalyte 148* comes in <u>two</u> formulation types:

1) Plasmalyte 148 ("Plain Plasmalyte") 1 Litre bags

Each <u>litre</u> contains:

Sodium	140	mmol/L
Chloride	98	mmol/L
Bicarbonate	27	mmol/L
Potassium	5	mmol/L
Magnesium	1.5	mmol/L

2) Plasmalyte 148 & Glucose 5% ("Plasmalyte and Five") 1 Litre bags

Same as above, but with 50grams of Glucose in each litre.

ADJUSTING THE GLUCOSE CONCENTRATION

To make Plasmalyte & **<u>5%</u>** (<u>only if "Plasmalyte and Five" is unavailable</u>):

Take a **<u>1 Litre</u>** bag of "Plain Plasmalyte": add **<u>110mL of Glucose 50%</u>** (55grams of glucose in 1110mL).

To make Plasmalyte & **<u>10%</u>** :

Take a **<u>1 Litre</u>** bag of "Plasmalyte and Five": add **<u>125mL of Glucose 50%</u>** (112.5 grams of glucose in 1125mL).

ADJUSTING THE POTASSIUM CONCENTRATION (for either type of Plasmalyte)**:**

To increase Potassium concentration to **<u>20mmol/L</u>**:

Take a **<u>1 Litre</u>** bag: add **<u>7.5mL of KCl 15%</u>** (15mmol of KCl to make total of 20mmol potassium per litre).

To increase Potassium concentration to **<u>40mmol/L</u>**:

Take a **<u>1 Litre</u>** bag: add **<u>17.5mL of KCl 15%</u>** (35mmol of KCl to make total of 40mmol potassium per litre).

* The "148" in "Plasmalyte 148" refers to the total equivalence of positive ions: 140 Na^+ + 5 mmol K^+ + 1.5mmol Mg^{2+} = 148

DRAFT

Authors: C Begg, N Wilson, A McGettrick, C Cairney
Version: 1.1 DRAFT
This document supports the RHC PICU IV Fluid Decision Support Tool & RHC PICU IV Fluid Therapy Guideline
Date of Issue: 19/05/2017
Date of Review: 19/05/2020